

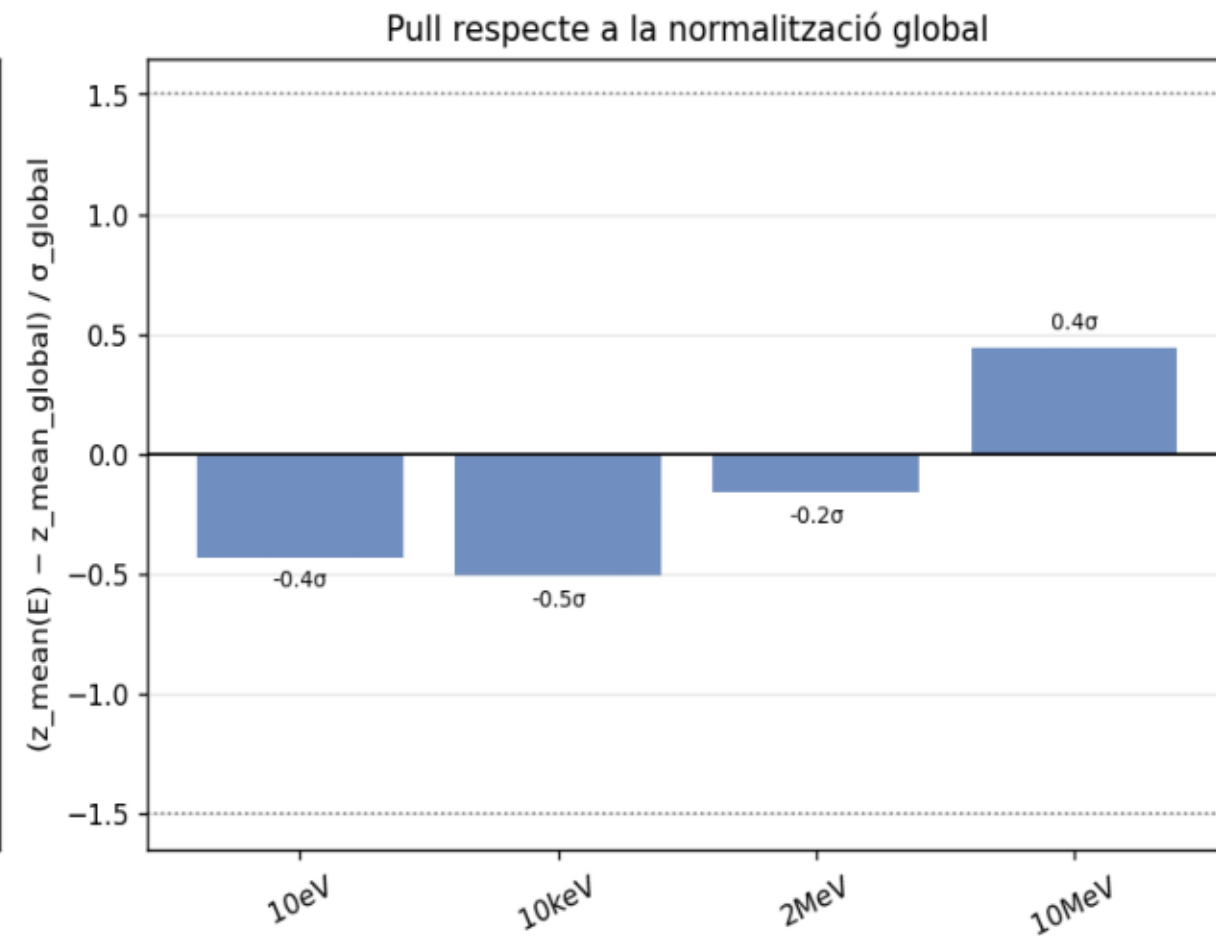
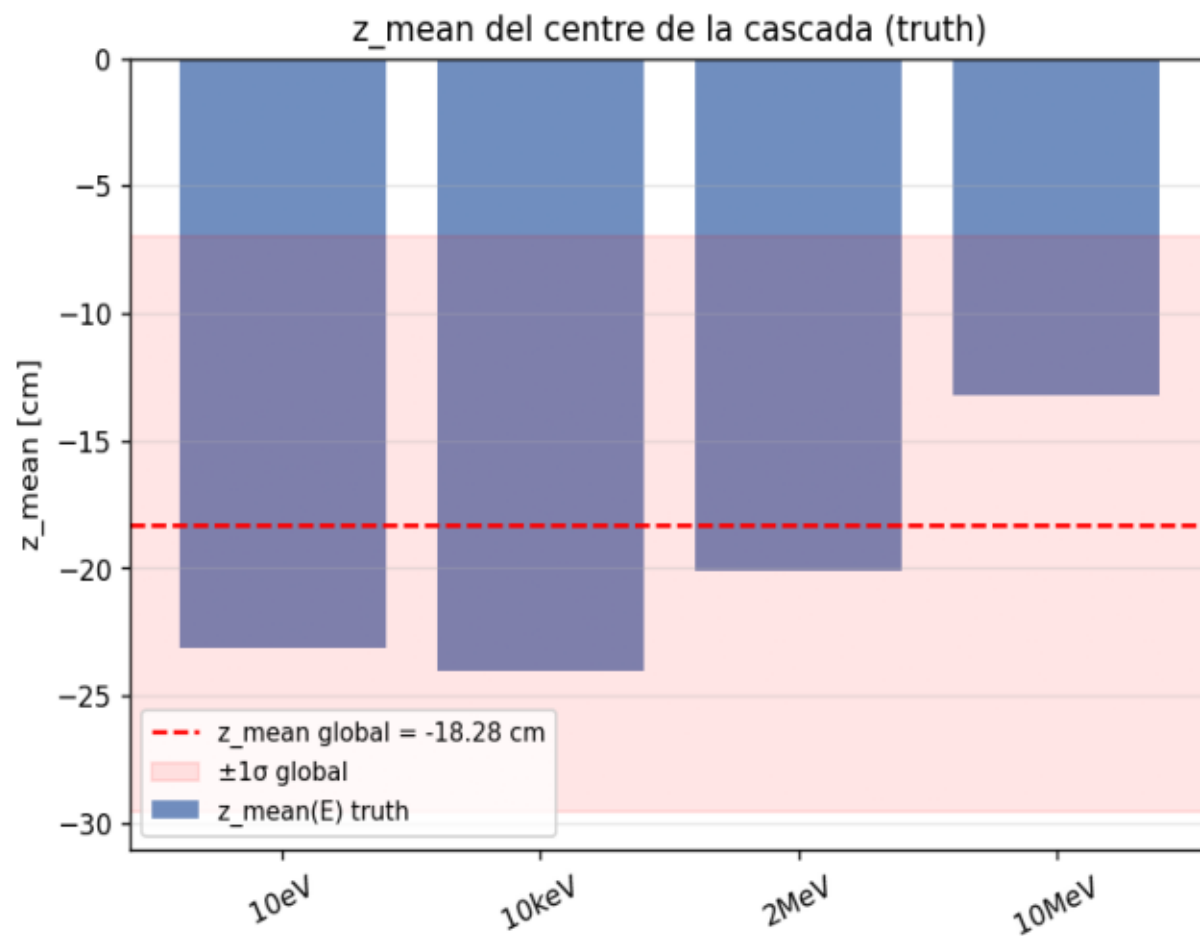
Part A — Paràmetres de normalització (transforms)

Dataset
(shared)

Paràmetre	Valor
z_mean	-18.052
z_std	10.603
R_max	20
log_edep_mean	-4.4352
log_edep_std	2.9554
edep_offset_MeV	1e-12
log_t_mean	3.6413
log_t_std	1.8725
t_offset_ns	0.001
log_E_in_mean	-2.2675
log_E_in_std	3.1847
z_mean_poly_coeffs	[0.1187, 1.4921, 4.8059, -20.731]
condZ	✓

B z per energy truth

Part B — z_mean per energia (truth) vs referència global

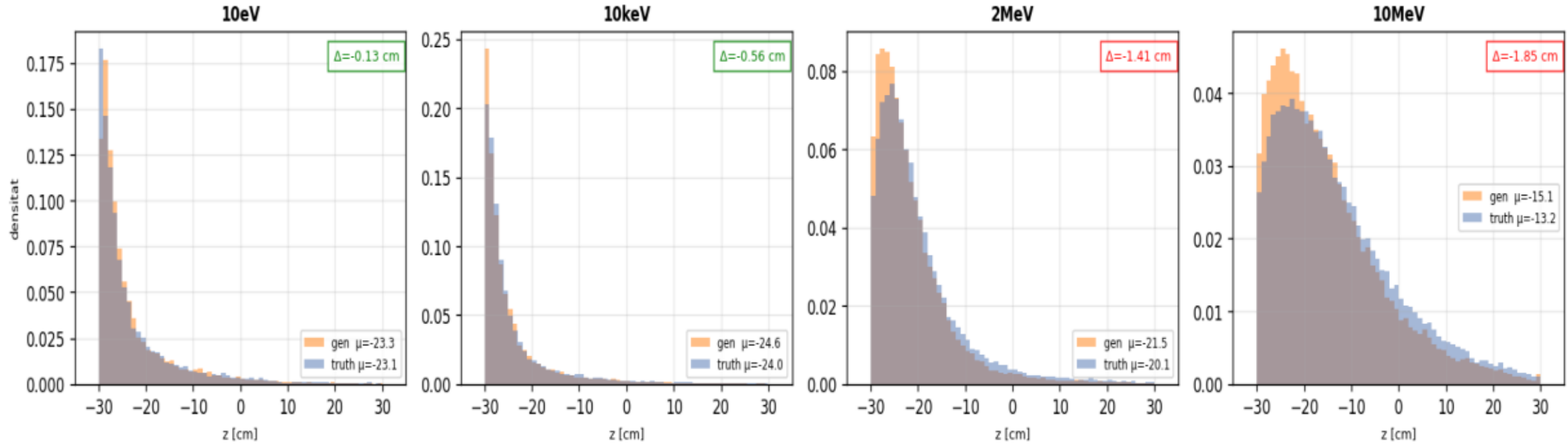


Dataset
(shared)

C z phys

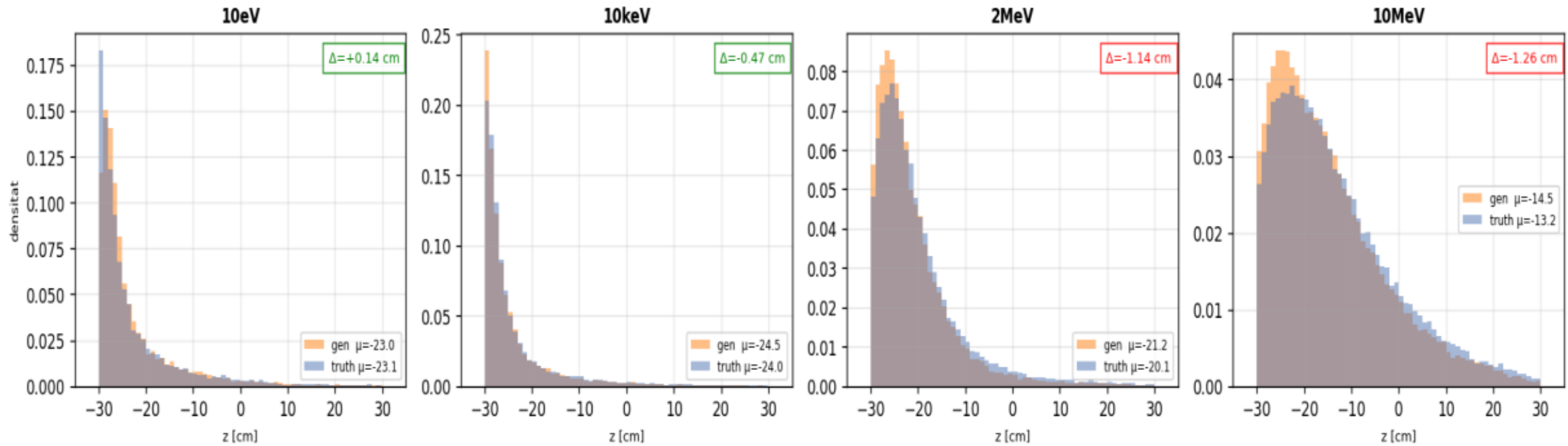
Part C — Distribució z física [cm] (truth vs generated)
Blau = truth (Geant4) Taronja = generated (EPiC-FM)

run_007 (fs=5)



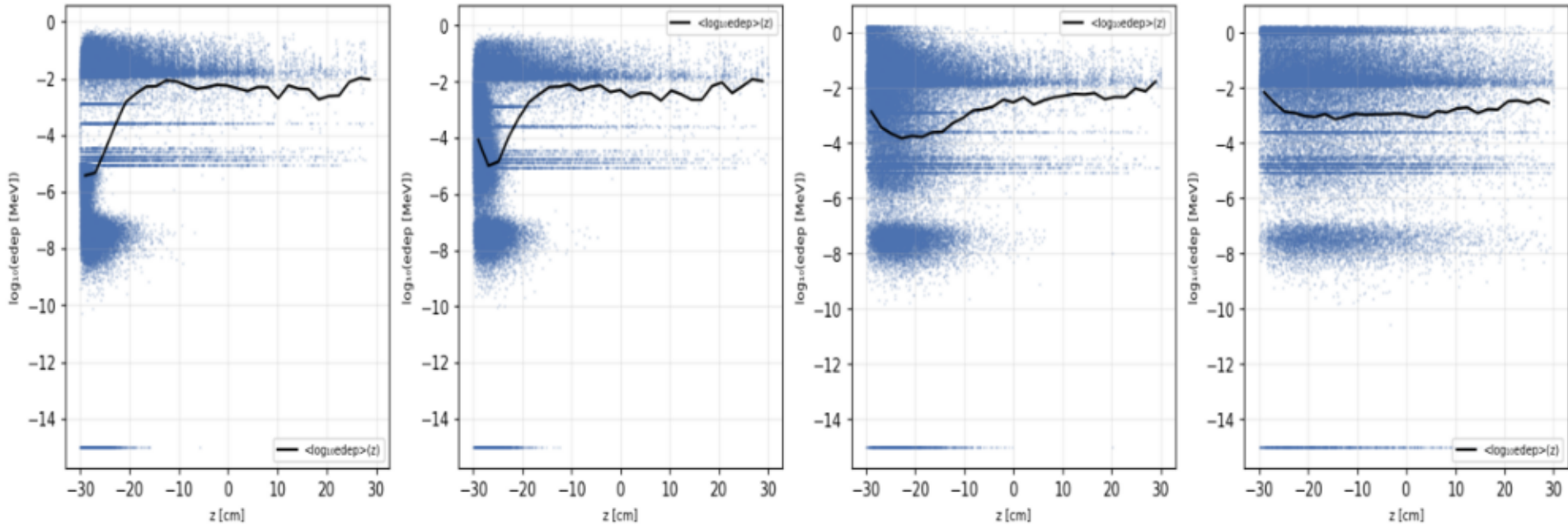
Part C — Distribució z física [cm] (truth vs generated)
Blau = truth (Geant4) Taronja = generated (EPiC-FM)

run_010 (fs=2)

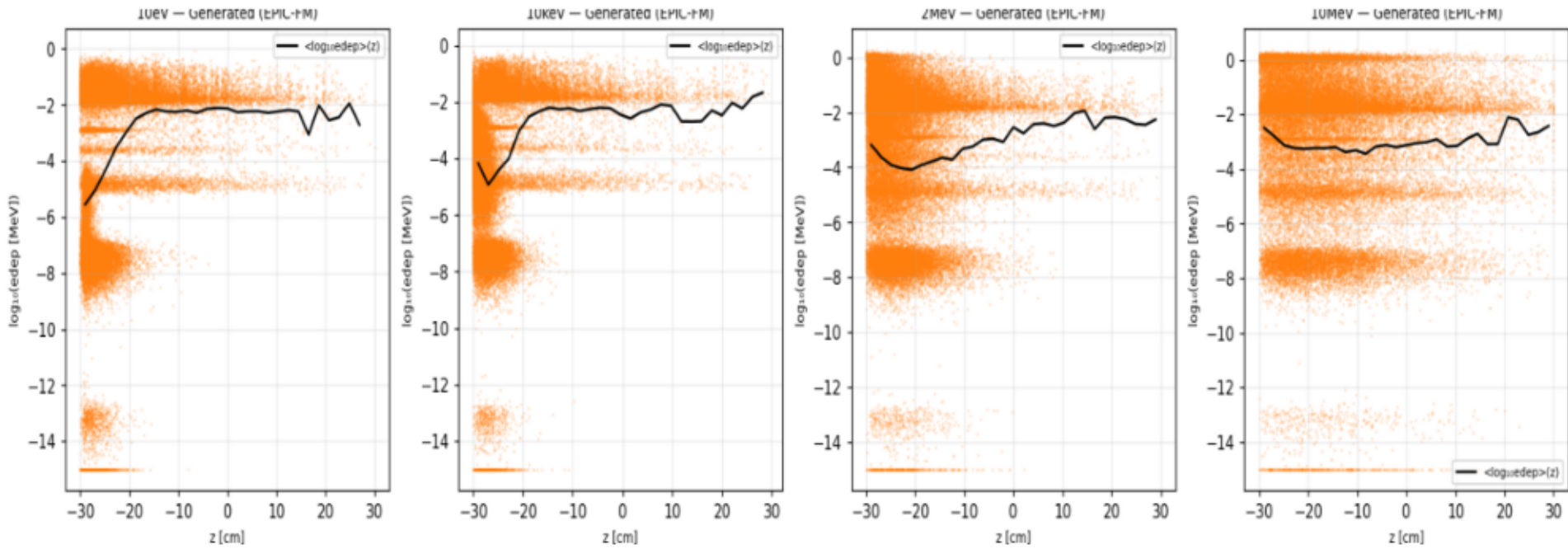


D edep vs z scatter

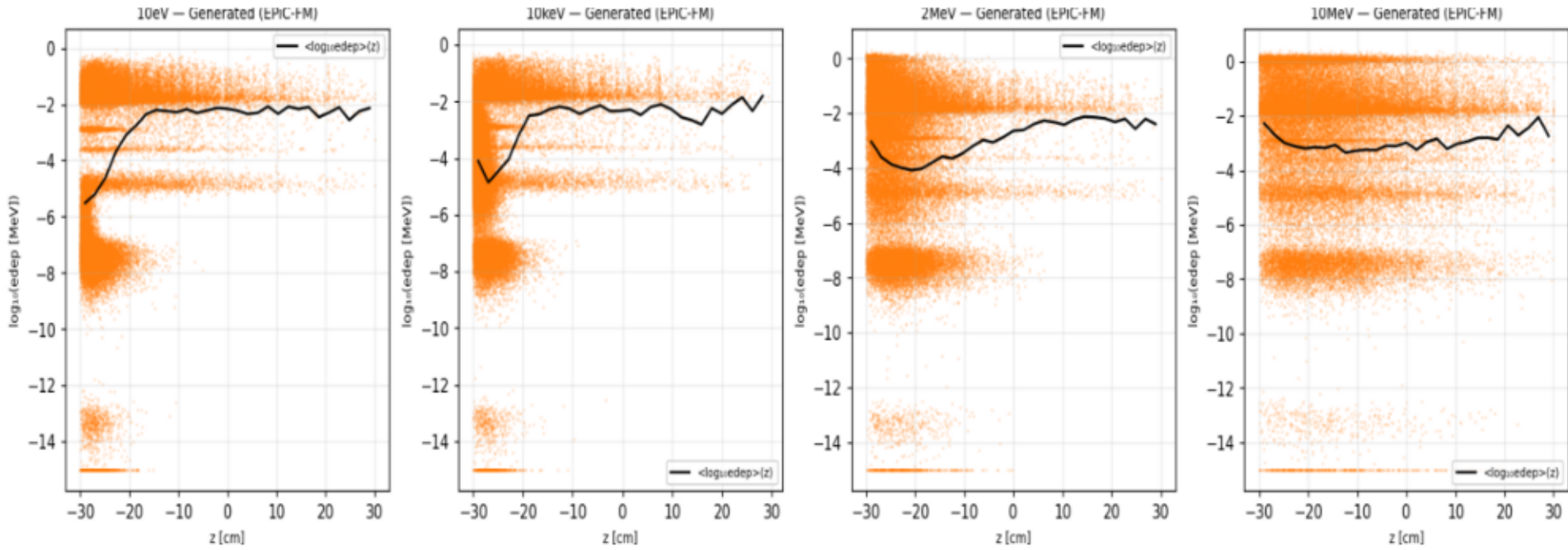
Truth (Geant4)



run_007 (fs=5)



run_010 (fs=2)

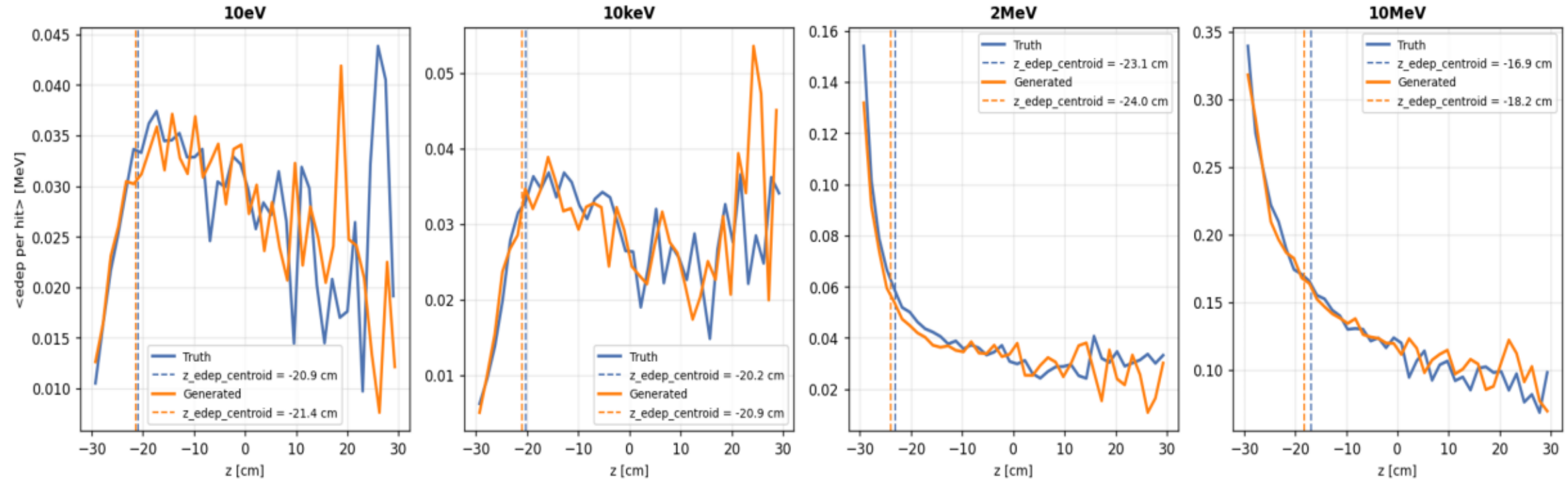


Bandes horitzontals esperades (neutrons en CH2):

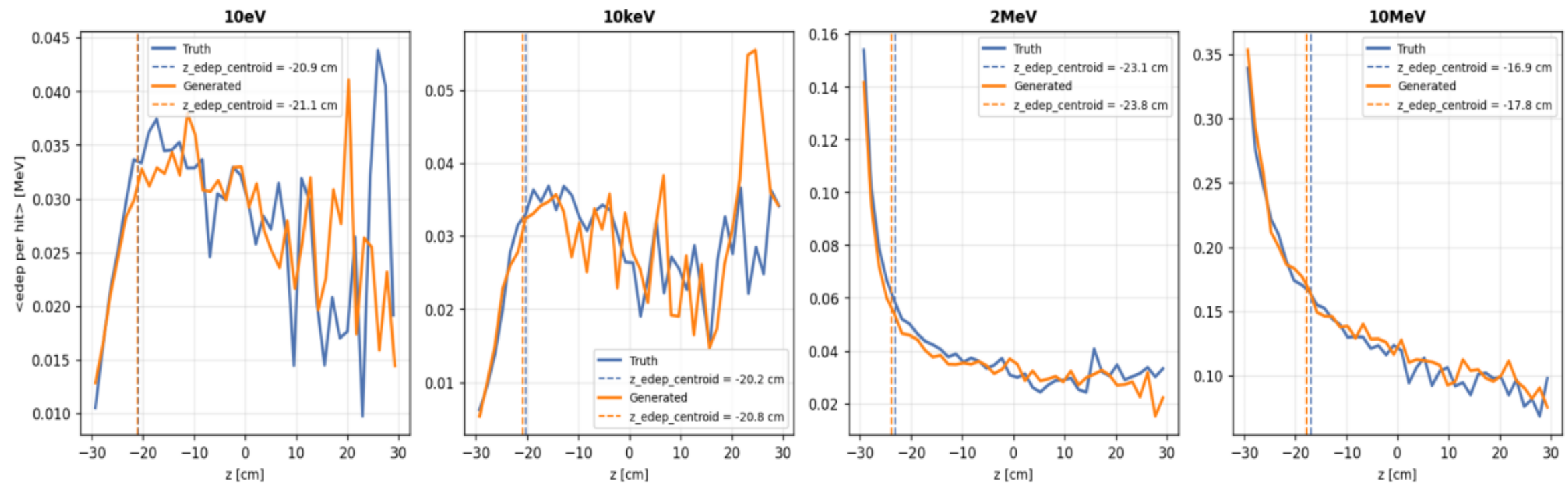
$\log_{10}(\text{edep}) \sim +0.65$	$^{12}\text{C}(n,n'\text{g})$ inelastic, $g = 4.44$ MeV
$\log_{10}(\text{edep}) \sim +0.35$	$\text{H}(n,g)\text{D}$ captura termica H, $g = 2.22$ MeV
$\log_{10}(\text{edep}) \sim +0.10$	$^{12}\text{C}(n,g)$ captura C, $g = 1.26$ MeV
continu	$n(\text{H},\text{H})n$ recul de proto (elastic), E in $[0, E_n]$
continu	Compton / fotoelectric de g secundaris
$\log_{10}(\text{edep}) \sim -14$	Floor de tracking Geant4 (electrons baixa E)

E edep z profile

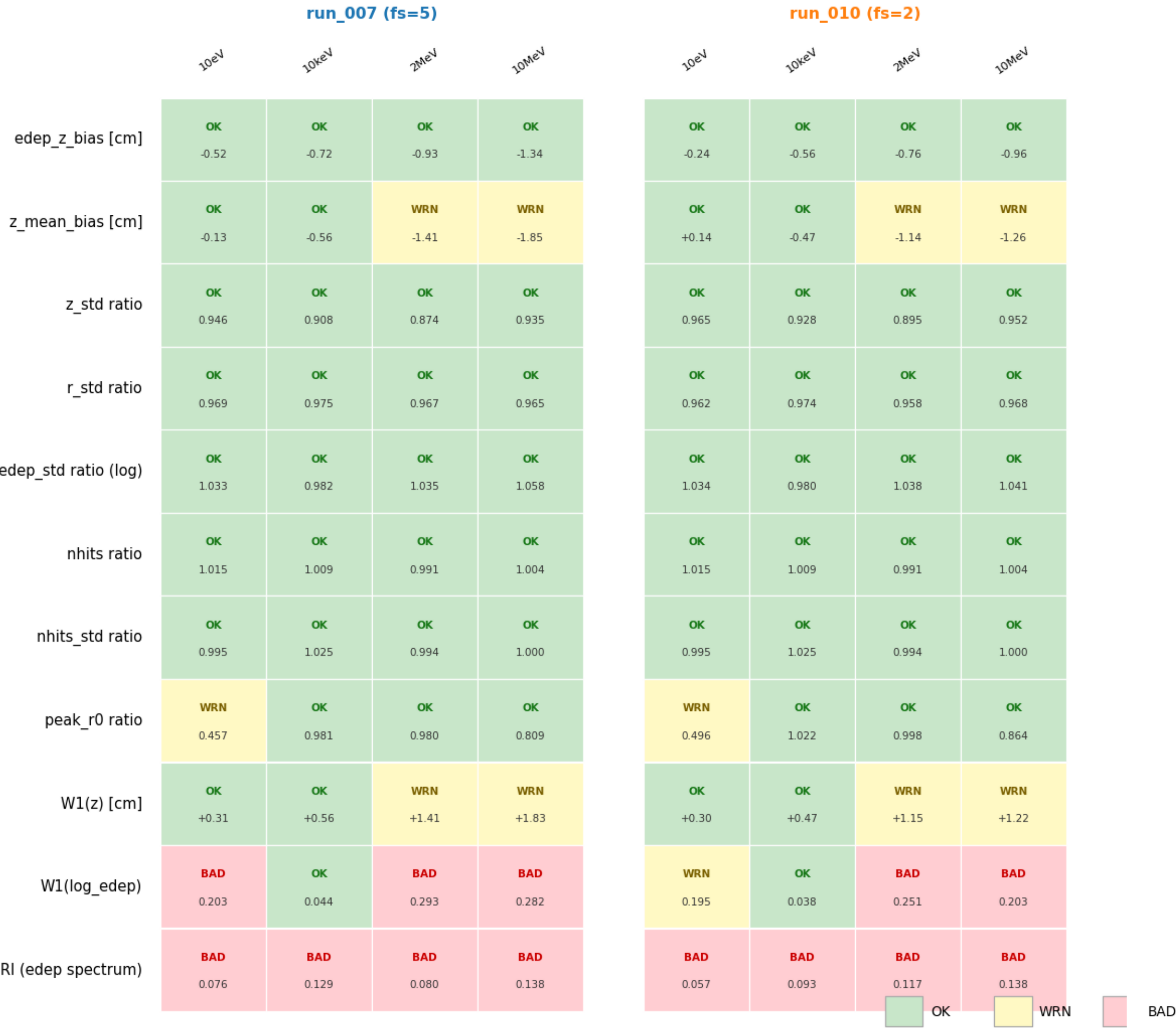
Part E — Perfil $\langle \text{edep} \rangle(z)$ binned
Blau = truth Taronja = generated (les línies verticals = centroide z)



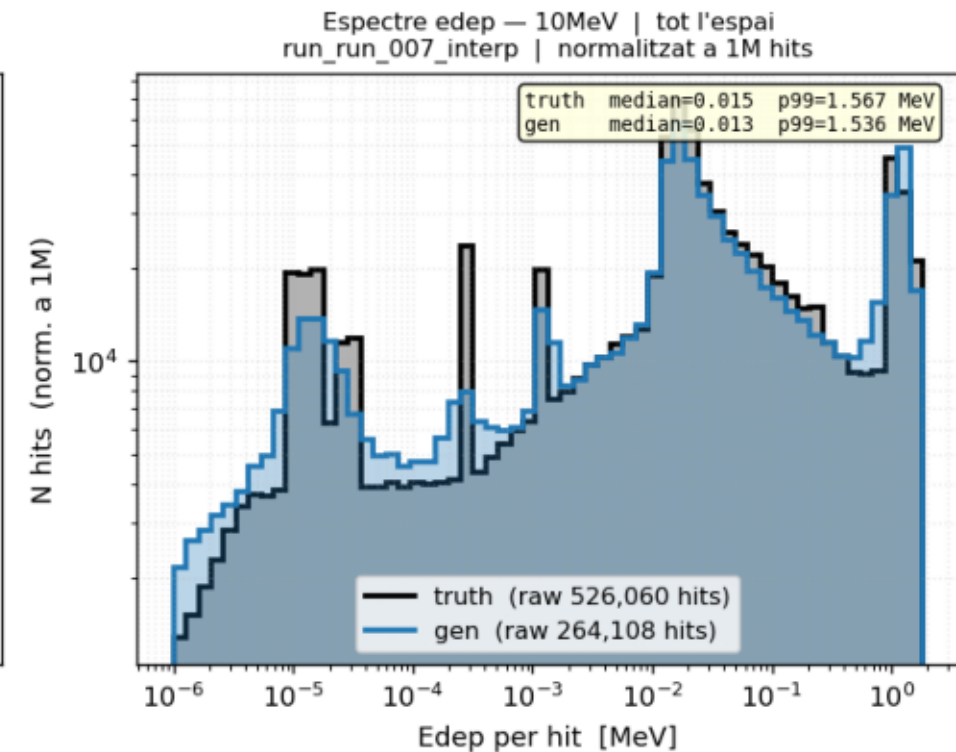
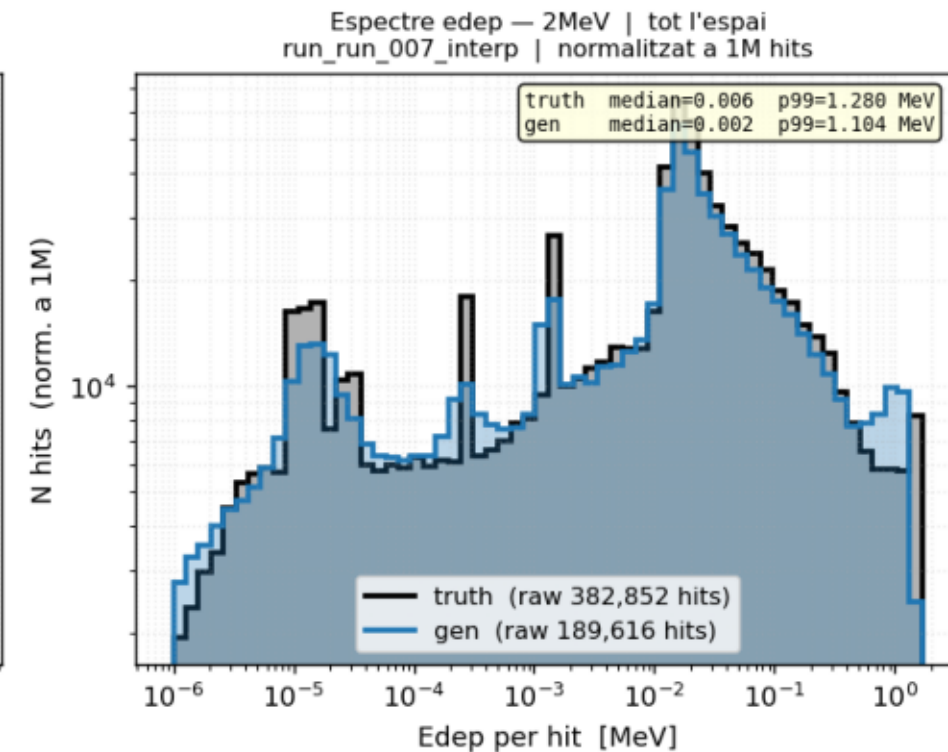
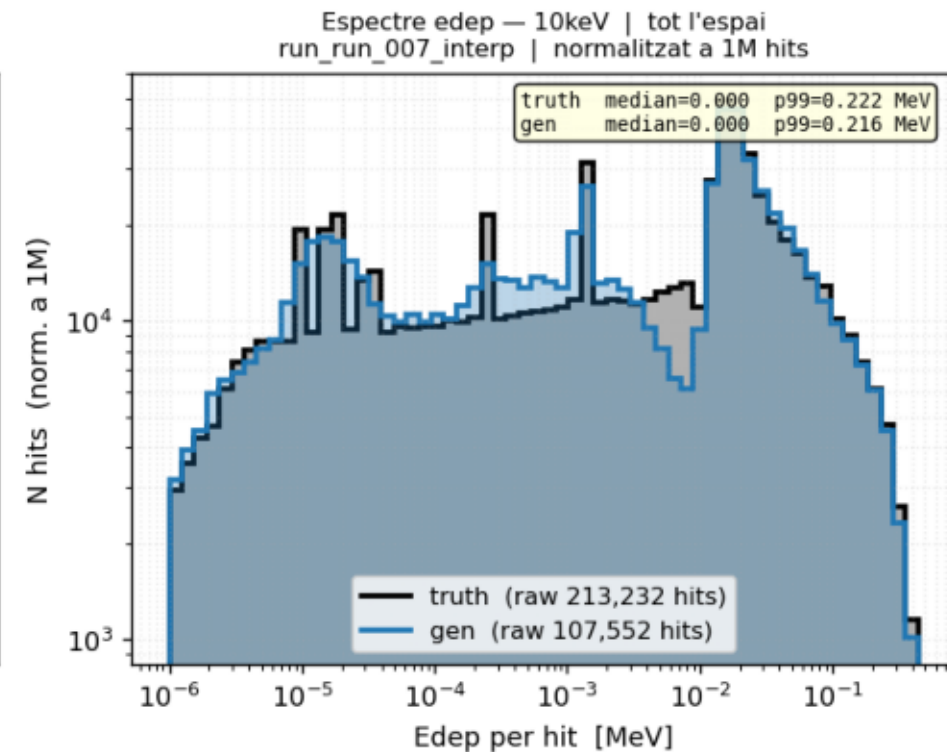
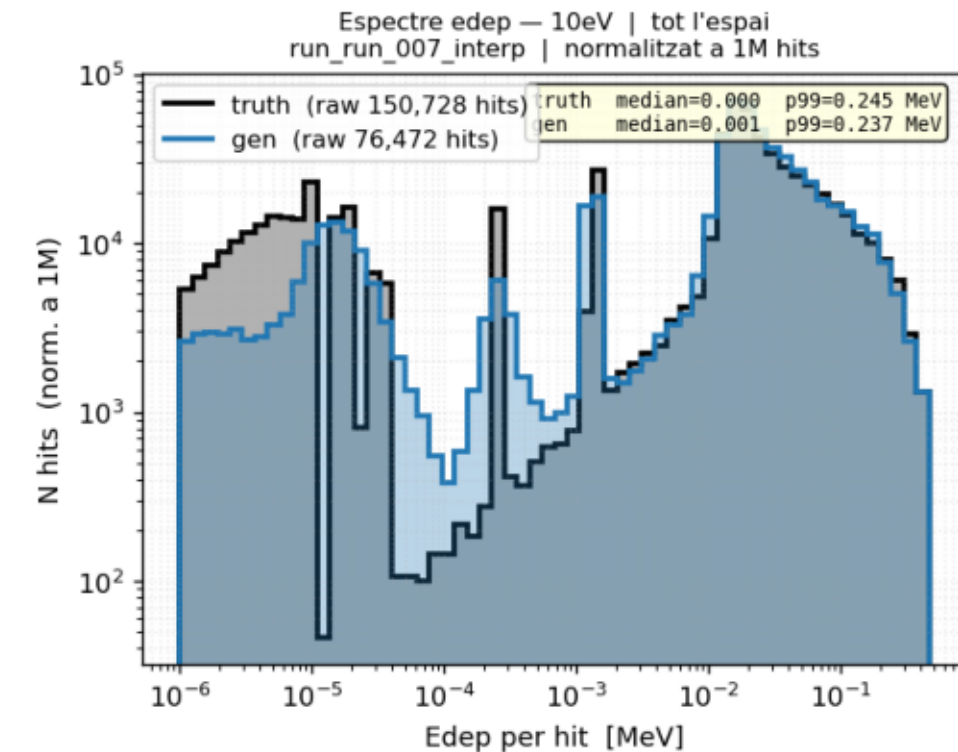
Part E — Perfil $\langle \text{edep} \rangle(z)$ binned
Blau = truth Taronja = generated (les línies verticals = centroide z)



Part Z - Heatmap metriques OK/WRN/BAD per run i energia



Mode=edep | run run_run_007_interp | z=[-30,30] cm cm



Mode=edep | run run_run_010_interp | z=[-30,30] cm cm

